

# Day 7: Image Color Spaces

## Outcomes:

- Understand what a **color space** is
- Know the difference between **BGR, Grayscale, and HSV**
- Convert images between color spaces using OpenCV
- Understand when and why different color spaces are used

# What is a color space?

A color space is a way to represent colors using numbers.

BGR and Grayscale are two already known color spaces.

## BGR color space

Shape:

```
(height, width, 3)
```

This is the default color space in OpenCV.

```
BGR → [Blue, Green, Red]
```

- Each channel ranges from 0-255

## Grayscale

Grayscale removes color and keeps intensity.

Converting to grayscale:

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

Shape:

```
[height, width]
```

- One channel only, ranges from 0-255 (8-bit images, default)
- 0 → black, 255 → white
- Values in between → shades of gray
- Used in edge detection, face detection, thresholding

## HSV color space (imp)

Converting into HSV:

```
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
```

H → Hue

S → Saturation

V → Value

HSV separates **color** from **brightness**, which is extremely useful.

## Why HSV

- In BGR, Color and brightness are mixed.
- In HSV, you can detect colors regardless of lighting.
- Hence, HSV is used in color detection, object tracking, segmentation.

## Viewing HSV channels separately

```
h, s, v = cv2.split(hsv)

cv2.imshow("Hue", h)
cv2.imshow("Saturation", s)
cv2.imshow("Value", v)
```

12-hsv.py

```
import cv2 as cv

img = cv.imread("OpenCV-codes/images/image.png")
hsv = cv.cvtColor(img, cv.COLOR_BGR2HSV)

h, s, v = cv.split(hsv)

cv.imshow("Hue",h)
cv.imshow("Saturation", s)
cv.imshow("Value", v)

cv.waitKey()
cv.destroyAllWindows()
```

# Separated H, S, V View

What a separated h, s, v look like as an image (refer 12-hsv.py)

## Hue

- Looks black and white
- Stores color type as numbers ranging from 0 – 179
- It does not represent brightness

## Saturation

- Looks black and white and ranges from 0 – 255
  - 0 (black when separated) → no color (gray)
  - 255 (white when separated) → strong color
- Controls **color intensity**, not brightness

## Value

- Controls light/dark
- No effect on color type

Note: If S = 0 (no color) → grayscale, even if hue has a value, it becomes meaningless.

## Conclusion

BGR – good for display

Grayscale – good for processing

HSV – good for color-based processing.